

AMAL P K

Cloud & DevOps Engineer

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PROFESSIONAL SUMMARY

Cloud and DevOps Engineer with hands-on experience in AWS, CI/CD, Infrastructure as Code, and containerized deployments. Skilled in Terraform, Ansible, Docker, Kubernetes, cloud networking, and automation. Background in Python and Node.js backend development, supporting end-to-end application deployment and operations.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (EC2, EKS, ECS, S3, RDS, ALB, IAM, VPC, ACM)

Containers & Orchestration: Docker, Docker Compose, Kubernetes, Helm, Amazon EKS

CI/CD & Automation: GitHub Actions, Jenkins, Ansible, Terraform, Shell Scripting

Monitoring & Web Servers: CloudWatch, Prometheus, Grafana, Nginx, ALB

Backend & Scripting: Python, Node.js, RESTful API Design

Databases: PostgreSQL, MySQL, MongoDB

Tools & Platforms: Git, GitHub, Linux, Postman

WORK EXPERIENCE

AWS DevOps Engineer Trainee | Dec 2025 – Present

Cokonet Academy, Trivandrum

- Provisioned and maintained cloud infrastructure on AWS (EC2, S3, IAM, VPC, EKS) following security best practices including least-privilege IAM policies and private-subnet isolation.
- Built multi-stage CI/CD pipelines using Jenkins and GitHub Actions, reducing manual deployment effort significantly.
- Containerised applications with Docker and orchestrated deployments on Kubernetes (EKS), ensuring rolling updates and zero-downtime releases.
- Automated infrastructure provisioning with Terraform and Ansible, enabling repeatable, version-controlled environment creation.
- Configured Nginx as a reverse proxy and load balancer for containerised workloads in production environments.

Junior Backend Developer | Jun 2025 – Dec 2025

Adviciya Solutions, Ernakulam

- Designed and developed RESTful APIs in Python, implementing core business logic and optimising PostgreSQL queries for faster response times.
- Integrated GCP Cloud Tasks for distributed background job scheduling and Firebase Cloud Messaging (FCM) for push notifications.
- Managed IAM roles, VPC configurations, and network security rules across cloud environments.
- Collaborated in Git-based workflows with feature branching, code review via pull requests, and structured release management.

Full Stack Web Development Intern | Jan 2025 – Mar 2025

Edunet Foundation (EY GDS & AICTE)

- Completed a 6-week internship focused on MERN stack development and full-stack application architecture.
- Built a personal finance tracker with CRUD operations, JWT/bcrypt authentication, and Chart.js dashboards.
- Developed and integrated RESTful APIs connecting React frontend to Node.js/Express backend.

PROJECTS

Dockerized MERN Application | GitHub

- Containerized a full-stack MERN app with Docker Compose, orchestrating frontend, backend, and MongoDB with health-checked startup.
- Built a multi-stage Docker image for the React frontend (Node Alpine build → Nginx Alpine runtime), keeping Node and npm out of the final image.
- Configured Nginx as a reverse proxy, serving static assets and routing API requests through a single entry point.
- Tech: Docker, Docker Compose, Nginx, React, Node.js, Express.js, MongoDB, AWS EC2.

CI/CD Pipeline(Spring Boot)- GitHub Actions, Docker & EC2 | GitHub

- Built a GitHub Actions pipeline that runs the Maven test suite, builds a multi-stage Docker image (Maven/JDK build stage → JRE-Alpine runtime stage), and pushes to Amazon ECR on every push to main, tagged with both the Git commit SHA and latest for traceability.
- Replaced static AWS access keys with IAM OIDC federation, letting GitHub Actions assume a scoped IAM role for short-lived credentials instead of storing long-lived secrets.
- Automated deployment to EC2 over SSH: the pipeline pulls the new image, replaces the running container, and verifies the rollout with a post-deploy health check against /health before marking the job successful.
- Reduced production image size by separating build and runtime stages — compiling with the full Maven/JDK toolchain, then shipping only the compiled JAR on a minimal JRE-Alpine base.
- [Tech: Java, Spring Boot, Maven, GitHub Actions, Docker, Amazon ECR, Amazon EC2, AWS IAM OIDC.](#)

Multi-Tier AWS Infrastructure with Terraform | GitHub

- Provisioned a production-grade three-tier VPC (public, private, database subnets) across two AZs using reusable Terraform modules.
- Automated creation of ALB, RDS (PostgreSQL), and IAM roles — full stack in minutes.
- Stored Terraform state in S3 backend with DynamoDB locking for safe team collaboration.
- [Tech: Terraform, AWS VPC, EC2, ALB, RDS, S3, IAM.](#)

Amazon EKS Deployment | GitHub

- Deployed containerised application on Amazon EKS with node groups, namespaces, Deployments, Services, and Ingress resources.
- Set up AWS Load Balancer Controller and ALB Ingress for external traffic with SSL via ACM.
- Implemented Horizontal Pod Autoscaler (HPA) to scale pods on CPU utilisation for high availability.
- [Tech: Amazon EKS, Kubernetes, Helm, ALB Ingress Controller, HPA, ACM, IAM OIDC.](#)

Prometheus & Grafana Monitoring on Kubernetes | GitHub

- Deployed kube-prometheus-stack via Helm on EKS, collecting metrics from all nodes, pods, and control-plane components.
- Built custom Grafana dashboards tracking pod CPU/memory, request latency, and node health with configurable thresholds.
- Configured Prometheus alerting rules and routed critical alerts to Slack via Alertmanager webhooks.
- [Tech: Kubernetes \(EKS\), Helm, Prometheus, Grafana, Alertmanager, Slack.](#)

EDUCATION

B.Tech in Information Technology | APJ Abdul Kalam Technological University, Thalassery | CGPA: 7.5

Diploma in Engineering (Wood and Paper Technology) | Govt Polytechnic College, Kannur | CGPA: 8.6

Higher Secondary Education (Biology Science) | Kerala Board | 70%